Ignytion io

IGNYTION IO · TECHNICAL LAB NOTE

Compile a Sobel Kernel from C to SKY130

Actual Bambu RTL, APB integration, routed timing, and independent layout checks

A small image-gradient kernel provides a traceable path from C behaviour to a routed digital core. PandA Bambu generates the arithmetic RTL. An APB wrapper supplies pixels and captures the result. Independent numerical checks, gate simulations, extracted timing, Magic DRC, and Netgen LVS test the implementation at their stated scope.

SECTION 01

What the local result establishes

The **C kernel, generated core, and APB-controlled implementation** agree on 5,162 frozen test windows. The routed candidate also meets its declared nominal timing and electrical checks.

CHECK	MEASURED RESULT	STATE
C and generated RTL	5,162 directed/deterministic windows	✓ PASS
APB and gate netlists	67,124 checks per APB run	✓ PASS
TT extracted setup / hold	+11.72 / +0.12 ns at 20 ns	✓ PASS
Slew / capacitance / fanout	Zero reported violations	✓ PASS
Magic / hierarchical LVS	0 DRC errors · 874 devices / 921 nets match	✓ PASS
Negative controls	Wrong result, broken net, and 1 ns target rejected	✓ PASS

Crucible is beta. It supplies the versioned tool environment for the local flow. The SKY130 PDK and native host/runtime dependencies remain explicit prerequisites.

A bounded engineering result

Vector agreement is sampled functional evidence. Timing covers **TT, 25 °C, 1.80 V and nominal extracted RC**. This result does not establish exhaustive equivalence, other corners, physical FPGA operation, or manufacturing signoff.

SECTION 02

Freeze the numerical contract

Each request supplies a complete **3 × 3 window of unsigned eight-bit pixels**. The centre pixel is p4; the input bytes run from p0 to p8 in row-major order.

WINDOW ROW	GX COEFFICIENTS	GY COEFFICIENTS
p0, p1, p2	−1, 0, +1	−1, −2, −1
p3, p4, p5	−2, 0, +2	0, 0, 0
p6, p7, p8	−1, 0, +1	+1, +2, +1

The output is **min(255, |gx| + |gy|)**. Signed C integer intermediates hold each gradient, whose magnitude is at most 1,020 for eight-bit pixels. The result is saturated before conversion to an unsigned byte.

This is the **L1 gradient approximation**. It does not compute a square root. The caller supplies all nine pixels, so image-border handling, line buffering, frame storage and image transport remain outside this window kernel.

> **Type identity was checked**
> The freestanding interface uses built-in unsigned char rather than a libc typedef. The compiled C reference is compared independently, the frozen vector hash is enforced, and the actual generated input/output ports are checked to be eight bits wide.

SECTION 03

Inspect what Bambu generated

PandA Bambu 2024.10 compiled the actual C kernel to Verilog. The generated file retains its compiler revision and IP-library notices; its arithmetic is used directly in the implementation.

```
cd build/hls
igny run --env hls-sobel bambu ../../src/sobel.c \
  --top-fname=sobel3x3 --clock-period=20
```

PORT OR PROPERTY	OBSERVED GENERATION
p0 through p8	Nine unsigned 8-bit inputs
return_port	Unsigned 8-bit output
reset	Synchronous, active low
start_port / done_port	Done follows start in the generated single-state controller
Arithmetic datapath	Combinational; Bambu reports zero datapath flip-flops

The **core adapter only maps ports and byte order**. It does not reimplement Sobel arithmetic. The core test observes completion at its first sampled edge after applying start and compares the result with the frozen byte.

That handshake observation is not a general promise about HLS latency. This particular generated controller has one state; a different kernel or Bambu configuration may produce a multi-cycle interface.

SECTION 04

Make the result accessible over APB

The wrapper owns **pixel storage, start, busy, sticky done and the captured result**. It uses one clock and an active-low reset, with zero-wait APB accesses.

OFFSET	REGISTER	CONTRACT
0x00	Control	Write bit 0 to start; read returns zero
0x04	Status	Read bit 0 = busy; bit 1 = done
0x08	Result	Read the saturated result in bits 7:0
0x10	Pixels 0–3	Four bytes; p0 is the least significant byte
0x14	Pixels 4–7	Four bytes; p4 is the least significant byte
0x18	Pixel 8	p8 is in bits 7:0; all 32 bits read back

A completed start write sets **busy** and clears done. The following core-completion edge captures the output, clears busy, and sets done. Reset clears all storage. Setup phases have no write side effects.

Unmapped or unaligned addresses and writes to status/result return an error. Busy writes are guarded in the wrapper, but the combinational core finishes before the next legal APB access; the suite does not claim a multi-cycle backpressure demonstration.

SECTION 05

Use an independent numerical observer

A Python matrix-kernel calculation is compared with the **compiled C function** before the resulting windows and expected bytes are frozen for RTL simulation.

VECTOR CLASS	COUNT OR OBSERVATION
Directed constant / impulse / edge cases	42 windows
Low-range deterministic random pixels	4,096 windows · values 0–31
Full-byte deterministic random pixels	1,024 windows · values 0–255
Output diversity	126 distinct result bytes
Unsaturated results	4,382 windows
Frozen seed	0x50BE1

The low-range cases exercise arithmetic below the saturation limit. Constant, impulse and directional-edge cases test zero response, pixel position and gradient orientation. Both positive and negative gradients occur.

The **same frozen vectors** drive the generated core, APB wrapper, mapped netlist and routed netlist. Four-state comparisons reject unknown results. Watchdogs convert a stalled test into a fatal error.

```
SOBEL_GENERATED_CORE_PASS vectors=5162 max_cycles=1
SOBEL_APB_PASS vectors=5162 checks=67124 max_polls=1
```

SECTION 06

Require the observers to reject faults

A positive run is paired with **deliberate, causal failures**. Golden RTL, constraints and layout remain intact while scratch inputs or comparison references are corrupted.

CONTROL	EXPECTED OBSERVATION	MEASURED OUTCOME
Wrong expected byte	Zero window should return zero	Expected 1, observed 0; exit 1
Disconnected result-Q net	Disconnect differs from layout	Netgen reports netlists do not match
1 ns clock constraint	Route misses the target	-7.2826 ns setup slack; violated

The runner checks both a **nonzero simulator exit and the causal mismatch** for the corrupted byte. LVS success is read from the comparison result, because Netgen process exit alone does not establish a match.

The timing control reuses the routed OpenDB and extracted SPEF, then replaces only the analysis clock with 1 ns. It leaves the implemented design and saved 20 ns SDC unchanged.

> **Limits of these controls**
> The controls show that these observers can discriminate the injected faults. They do not measure structural fault coverage or prove that every possible design bug would be detected.

SECTION 07

Implement a real routed core

Yosys maps the unchanged arithmetic and APB RTL to SKY130 HD cells. OpenROAD performs floorplanning, taps, PDN, placement, electrical repair, CTS, hold repair and routing.

ACTUAL GDS GEOMETRY

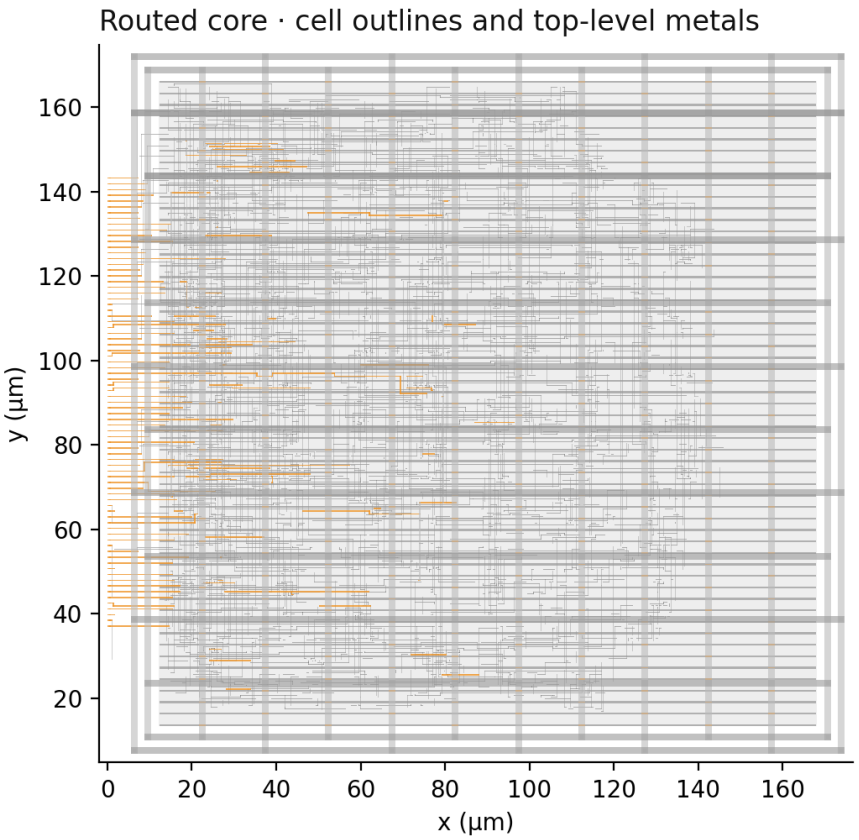


Figure 1 · Top-level metal geometry and instance outlines decoded from the streamed GDS. Metal 3 is orange; other routing and power-grid geometry are grey. The DEF die target is **180 × 180 µm**.

The GDS stream uses actual standard-cell mask views and an explicit SKY130 layer map. Supply ports carry names on metal 5, so independent extraction can compare VPWR and VGND as top-level pins.

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SECTION 08

Check timing and electrical limits together

OpenRCX SPEF and propagated clocks feed the final nominal analysis. Meeting setup and hold is accompanied by explicit slew, capacitance and fanout checks.

ASSUMPTION OR RESULT	VALUE
Clock period / uncertainty	20 ns / 0.1 ns
Data input delay · maximum / minimum	2.0 ns / 0.2 ns
Data input transition / output load	0.1 ns / 0.005 pF
Fanout limit	32
Library / extraction	TT 25 °C, 1.80 V / nominal OpenRCX rules
Extracted setup / hold worst slack	+11.72 ns / +0.12 ns
Slew, capacitance, fanout violations	0 reported

Placement-based repair first left one high-fanout address-select net above its routed electrical limit. A **30% slew/capacitance repair margin** and 0.1 ns hold-planning margin address the measured estimate-to-route shift. Final checks use extracted results.

Reset is explicitly false-pathed. Its **recovery/removal and distribution** remain outside this timing claim. Other PVT corners, RC corners, antenna, density, IR drop and electromigration require separate analysis.

SECTION 09

Compare independent layout evidence

Magic reads the **streamed GDS**, runs the installed SKY130 rule checks, and extracts connectivity. Netgen compares that extraction with the routed reference netlist.

EVIDENCE	OBSERVATION
TritonRoute report	Present and empty · routing checker scope
Magic DRC	Total DRC errors found: 0
Hierarchical Netgen match	874 devices and 921 nets on each side
Top-level supply pins	VPWR and VGND match by name
Disconnected result output	Netlists do not match

The comparison **retains power and body pins**. Physical filler/tap classes are exempted by the PDK setup. This is a hierarchical standard-cell connectivity comparison; it is not a flat transistor-level proof of every library cell.

The independent checks cover the actual GDS geometry, not only DEF generation. The tested tool versions and PDK files are recorded with hashes. Library geometry, extraction warnings and modelling limits remain part of the evidence.

> **No manufacturing qualification**
> Zero reported DRC errors and a unique LVS match are useful local evidence. Foundry acceptance, package/IO/ESD integration, complete signoff and silicon behaviour remain unestablished.

SECTION 10

Reproduce the staged flow

Use the **HLS_Sobel workspace in Linux/WSL**, its pinned Crucible tools, a complete SKY130 HD PDK, and a host C compiler/Python environment for the numerical reference.

```
igny init
igny workspace sync
bash scripts/reference.sh
bash scripts/run.sh hls
bash scripts/run.sh core
bash scripts/run.sh apb
bash scripts/run.sh synth
bash scripts/run.sh mapped
bash scripts/run.sh physical
bash scripts/run.sh stream
bash scripts/run.sh routed
bash scripts/physical_verify.sh
bash scripts/controls.sh
python3 scripts/verify.py
```

Set **PDKPATH** if the PDK is outside the default Ciel location. KLayout Python bindings are needed by the final observer. Bambu's packaged archive is approximately 1.39 GB; workspace sync may acquire it when absent. Native runtime dependencies must also resolve.

Builds and raw logs are regenerated locally. The repository keeps compact measured summaries and the final whitepaper in docs/. Independent clean-machine reproduction has not been demonstrated.

SECTION 11

Scope, provenance and source notes

This note demonstrates **one window kernel and one APB wrapper**. Its measured implementation is a nominal SKY130 digital-core candidate.

It does not include image line buffers, frame memory, DMA, video timing, exhaustive APB conformance, exhaustive numerical equivalence, SDF simulation, multiple clock domains, other PVT/RC corners, or FPGA board verification. GDS generation and local checks do not establish tapeout readiness.

PRIMARY TOOL AND PDK REFERENCES

PandA Bambu · C-to-Verilog tutorial
`https://panda.deib.polimi.it/?page_id=555`

OpenROAD · Gate Resizer
`https://openroad.readthedocs.io/en/latest/main/src/rsz/README.html`

SKY130 · foundry-provided standard-cell libraries
`https://skywater-pdk.readthedocs.io/en/main/contents/libraries/foundry-provided.html`

Source, managed workflow and recorded evidence
`https://github.com/CRUCIBLE-WORKBENCH/HLS_Sobel`

Measurements are from the local runs through **29 October 2026**. Bambu 2024.10, Yosys 0.63, OpenROAD 26Q3-771, Icarus 14, Magic 8.3.682, Netgen 1.5.323 and KLayout 0.30.10 retain their upstream credit. Exact artifact identities are in the project lock and evidence manifests. No endorsement is implied.